

Energy Transfer in Collisions

4th Grade • Language Supports for explaining with evidence

CONTENT GOAL

Design a fair marble investigation, change one variable, complete 3+ trials, record observations, analyze evidence, and explain what was learned.

Our science question

What happens when objects collide?



LANGUAGE GOAL

I can make a claim, use evidence from my observations, and explain how the evidence supports my claim.

Think → Sketch / home-language planning if helpful → Partner rehearse → Explain

Vocabulary for Our Investigation

Use the words while you investigate and explain — not just to memorize them.

COLLISION

when objects hit one another

✓ Two marbles hit.

Not: Marbles sit apart.

Sketch: Draw two circles touching + impact lines.

VARIABLE

one thing we change or test

✓ Change track height.

Not: Change height AND marble size.

Sketch: Circle the ONE thing that changes.

TRIAL

one time you do the test

✓ Trial 1, Trial 2, Trial 3

Not: Only talking about what might happen

Sketch: Draw 1 → 2 → 3 test boxes.

OBSERVATION

what you actually see or hear

✓ “The still marble moved.”

Not: “I think it will move.”

Sketch: Eye + ear beside a notes box.

TREND

a pattern found across data

✓ A similar motion change happens in 3 trials.

Not: One unusual result in one trial

Sketch: Repeat arrows across 3 rows.

EVIDENCE

observations/data used to support a claim

✓ Use results from your trials.

Not: A guess with no result

Sketch: Magnifying glass over a data table.

Student Talk Mat: Explain With Evidence

Rehearse with a partner before you write or record your final explanation.

1

CLAIM

I claim that when objects collide, _____.

2

EVIDENCE

In Trial ___, I observed _____.
Across our trials, _____.

3

REASONING

This evidence supports my claim because
_____.

WORD BANK

collision • variable • trial
observation • trend • evidence
motion • moved • stayed still
faster • slower • sound

LIGHTER STARTERS

I noticed _____.
My evidence is _____.
This shows _____ because
_____.

Partner move: Ask, “What did you observe that makes you think that?”

My own reasoning: _____

How to Build an Evidence-Based Explanation

Model uses an unrelated example so the collision answer stays yours.

UNRELATED TEST: Which paper towel absorbs more water?

CLAIM The thicker paper towel absorbed more water.

EVIDENCE In three trials, the thicker towel absorbed more water than the thinner towel.

REASONING The repeated trial results support the claim because the same pattern happened more than once.

Your turn: Use only what YOUR group actually observed. Do not invent a result.

Investigation Directions

One main action at a time. Keep the test fair.

1

CHOOSE one variable to change: marble size, track height, or release point.

2

KEEP the other factors the same so the investigation is fair.

3

WRITE clear steps another group could follow.

4

PRACTICE releasing and lining up the marbles before Trial 1.

5

TEST your setup at least 3 times.

6

RECORD what you actually see and hear in every trial.

7

LOOK for a trend — a pattern across your results.

8

EXPLAIN your claim with specific evidence and reasoning.

SELF-CHECK: ☐ one variable ☐ 3+ trials ☐ observations recorded ☐ trend checked ☐ claim + evidence + reasoning

Three Ways to Show What You Know

Same science thinking. Different ways to communicate it.

WRITE

Write a short claim-evidence-reasoning response.

Support: Word bank + sentence frames + sketch first

SPEAK

Give an oral explanation to the teacher/partner or record it.

Support: Talk mat + word bank + rehearsal

DRAW + EXPLAIN

Annotate a collision sketch/data display, then add a spoken or written explanation.

Support: Labels + arrows + word bank + sentence starters

SHARED SUCCESS CRITERIA

- ☐ I answer the testable question. ☐ I make a claim. ☐ I use specific evidence from my trials.
- ☐ I explain how the evidence supports my claim. ☐ I use science words accurately.

Language objective note: If students are required to demonstrate complete English sentences, the visual option still needs a brief oral or written explanation. Planning in a home language may be used before sharing.

Teacher Review Notes

Planning notes — not student-facing

Why these scaffolds?

Students already identify observations; these supports help them move from noticing → academic vocabulary → complete evidence-based explanation without lowering the investigation demand.

Fade the frames when...

students can independently name the claim, select relevant evidence, and connect it with reasoning. Move from full frames → open starters → no frame, while keeping the same success criteria.

SOURCE / CONTENT ACCURACY

Does every direction match the guide: one variable, 3+ trials, observations, trends, and evidence-supported claims?

PRESERVED THINKING DEMAND

Are students still designing, analyzing, and explaining — rather than only filling blanks or copying vocabulary?

LANGUAGE FIT

Are the words, sentence chunks, and visual supports enough for this class without over-scaffolding? Can students rehearse before responding?

Teacher assumption to review: the guide does not give a separate safety rule beyond careful setup/alignment and practice before Trial 1; add your classroom marble/track safety expectations before use.